



ULAB
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BANGLADESH

CSE2200: DESIGN PROJECT - II

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Project: AgroInvest – A Crowdfunding Platform for Farmers and Investors.

Introduction

AgroInvest is a web-based crowdfunding platform designed to connect farmers with investors through a secure and transparent online system. In Bangladesh, many farmers own agricultural land but often face financial difficulties in purchasing seeds, fertilizers, irrigation equipment, and other farming resources. Due to the lack of sufficient capital, many farming projects cannot be completed successfully. At the same time, many individuals are interested in investing small amounts of money but do not have a trusted platform to support agricultural projects.

The AgroInvest system aims to solve this problem by allowing verified farmers to create farming projects and request financial support from multiple investors. Investors can browse approved projects, invest through the SSLCommerz Sandbox payment gateway, and monitor funding progress and project milestones. The system also includes features such as farmer verification, trust score, investment history, and simulated profit/loss calculation to ensure transparency and improve user confidence.

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the AgroInvest platform. It serves as a guideline for the design, development, testing, and deployment of the system while ensuring that all stakeholder requirements are clearly defined and properly implemented.

2. Stakeholders & Users

Role	Description	Primary Goal
Farmer	Creates farming projects and requests investment.	Receive funding for crop cultivation and update project progress.
Investor	Invests in approved farming projects.	Invest securely and monitor project progress and expected returns.
Admin	Manages the entire system.	Verify farmers, approve projects, monitor investments, and maintain the platform.

3. Functional Requirements

These define the specific behaviors and functions the system must support.

3.1 Authentication Module

- **FR-01:** The system shall allow Farmers to register and log in using their email and password.
- **FR-02:** The system shall allow Investors to register and log in using their email and password.
- **FR-03:** The system shall allow the Admin to log in securely.
- **FR-04:** The system shall provide role-based access for Farmers, Investors, and Admin.

3.2 Farmer Module

- **FR-05:** Farmers can create new farming projects by providing crop details, required investment amount, project duration, and expected profit percentage.
- **FR-06:** Farmers can upload supporting documents (NID, land ownership document, and land image) for verification.
- **FR-07:** Farmers can edit or update project information before admin approval.
- **FR-08:** Farmers can update project milestones (e.g., Land Preparation, Planting, Harvesting).
- **FR-09:** Farmers can monitor the funding progress of their projects.

3.3 Investor Module

- **FR-10:** Investors can browse all approved farming projects.
- **FR-11:** Investors can view complete project details before investing.
- **FR-12:** Investors can invest in projects using SSLCommerz Sandbox.
- **FR-13:** Investors can view their investment history.
- **FR-14:** Investors can track the funding progress of each project.
- **FR-15:** Investors can view project milestone updates.
- **FR-16:** Investors can rate farmers after successful project completion.

- **FR-17:** The system shall display a **Farmer Trust Score** based on investor ratings, completed projects, and verification status.

3.4 Admin Module

- **FR-18:** Admin can verify farmer information and uploaded documents.
- **FR-19:** Admin can approve or reject farming projects.
- **FR-20:** Admin can manage Farmers and Investors.
- **FR-21:** Admin can monitor all investments and project status.
- **FR-22:** Admin can enter the final project profit or loss for simulation.
- **FR-23:** Admin can generate reports of projects and investments.

3.5 Investment & Payment Module

- **FR-24:** The system shall integrate SSLCommerz Sandbox for payment simulation.
- **FR-25:** The system shall record successful investment transactions.
- **FR-26:** The system shall automatically update the funding progress after each successful investment.
- **FR-27:** The system shall allow multiple investors to invest in a single farming project.

3.6 Profit/Loss Calculation Module

- **FR-28:** The system shall calculate the total profit or loss based on the final project result entered by the Admin.
- **FR-29:** The system shall calculate each investor's profit share according to their investment percentage.
- **FR-30:** The system shall display the calculated profit or loss for demonstration purposes only without performing real financial payout